

Aneesh Maganti

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EXPERIENCE

Affirm, New York City, NY, *Software Engineer, Traffic Engineering* July 2025 - Present

- Re-architected and built Affirm's secrets infrastructure end to end, including three-layer envelope encryption (KEK/DEK) with isolated IAM reader roles; shipped an engineer-facing decryption service on AWS Lambda with an agentic workflow, automating 40 requests/month and eliminating 60 hours/year of toil.
- Rewrote a core Kubernetes manifest build tool in Rust using agentic workflow, slashing Helm chart build times by 78% (from 14 minutes to 3 minutes); successfully drove team-wide adoption of the tool.
- Orchestrated the migration of External Secrets Operator (ESO) from v0.7.2 to v0.20.2 across 72 Kubernetes clusters, leveraging a Controller Class mechanism to run concurrent operators and manage diverse secret types.

Qualcomm, San Diego, CA, *Software Engineer Intern, AI Software/ML Group* May 2024 - Aug 2024

- Implemented various memory optimizations across the Qualcomm AI Runtime (QAIRT), reducing peak memory usage by 20% across benchmarked ResNet50 and LLAMA2 and LLAMA3 LLM models
- Optimized AIMET quantization pipeline via C++ PyBind API to reference static graph tensors within Python subprocess, reducing in-memory allocations by 50%
- Refactored QAIRT/AIMET toolchain via PyTorch 'meta' device to prevent unnecessary weight initialization and modified model export to use in-memory IR graph, reducing disk space and improving runtime

New York University, Brooklyn, NY, *Teaching Assistant (Machine Learning)* Sep 2023 - Dec 2023

- Instructed students weekly for machine learning topics of written and programming tasks through office hours
- Graded weekly assignments on the basis of proper algorithm implementation, code correctness and style

NYU Algorithms and Foundations Group, Brooklyn, NY, *ML Academic Researcher* Feb 2023 - Sep 2023

- Designed and implemented *Deltagonalshift*, a diagonal estimator for a dynamic matrix under Professor Christopher Musco to improve neural network optimization based on DeltaShift trace estimation algorithm
- Demonstrated Deltagonalshift was more effective than repeatedly running Hutchinson's diagonal estimator

Corelink, Brooklyn, NY, *Software Engineer Intern* Sep 2021 - May 2022

- Implemented a C++ UDP network packet splitter to enable researchers to bypass Corelink's MTU limit from 20,000 to 64,000 bytes, increasing maximum throughput by 220%
- Designed Next.js/React interview scheduling platform using Auth0 for authentication and MongoDB backend

EDUCATION

New York University, Tandon School of Engineering, Brooklyn, NY Fall 2024

Bachelor of Science, Computer Science

GPA: **3.80**

Relevant Courses: Algorithmic Machine Learning, Offensive Security, ML Visualization, Computer Architecture

PROJECTS/ACTIVITIES

Dot Matrix - Game Boy Emulator June 2025 - Present

- Re-architected my C++ Game Boy emulator into idiomatic Rust (CPU, MMU, PPU, APU, cartridge, joypad, and renderer modules), driving the system from a cycle-stepped fetch-execute loop with full interrupt handling and accurate timer/clock emulation, and verified CPU correctness against Blargg's cpu_instrs test ROM suite.
- Built a scanline-based PPU implementing the four-mode LCD state machine (OAM scan, pixel transfer, H-Blank, V-Blank) with background, window, and sprite rendering output through a minifb framebuffer.
- Added MBC1 ROM/RAM bank switching, input, audio, and save states, enabling playback of commercial titles.

BUGS Open Source Club President Sep 2022 - Dec 2023

- Started and led 50+ member club by coordinating biweekly workshop and project coding events to discuss software engineering skills, foster contributions to open source, and create a fun, inclusive CS community.
- Led multiple workshops including discussion of open-source licenses, JavaScript Playwright automation, and an overview of emulation and the internals of C++ Game Boy emulator.
- Developed Next.js-React website NYU Syllabi with Netlify hosting and Docusaurus-based NYU CS Wiki websites